Statistical Inference

POLS 602
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Today

- A quick review of probability theory
- What are we estimating, and how do we quantify uncertainty about it?
- How do we test whether a coefficient, or a whole model, is different from zero?
- How do we interpret statistical tests well?

Probability Review

Why Probability?

- Finite samples are random, so our estimates are uncertain
- Probability gives us a language for that uncertainty
- Probability describes random error, not bias
- Large samples and confidence intervals do not fix bias

Two Definitions of Probability

- **Frequentist**: the proportion of times an event occurs over infinite identical trials
- **Bayesian**: a subjective belief about the relative likelihood of an event

This course deals in frequentist statistics.

Key Terms

Trials, Outcomes, and Events

- **Trial**: an action that produces an outcome of interest, such as rolling a die
- **Outcome**: the result of a trial, such as rolling a 3
- **Event**: a set of outcomes, such as rolling a number less than 3

Sample Spaces and Random Variables

- **Mutually exclusive events**: events that share no outcomes
- **Sample space**: the set of all possible outcomes of a trial, written Ω
- **Random variable**: a number assigned to each outcome of a trial

Probability Distributions

Probability Distribution

A probability distribution gives the likelihood of each value a random variable can take.

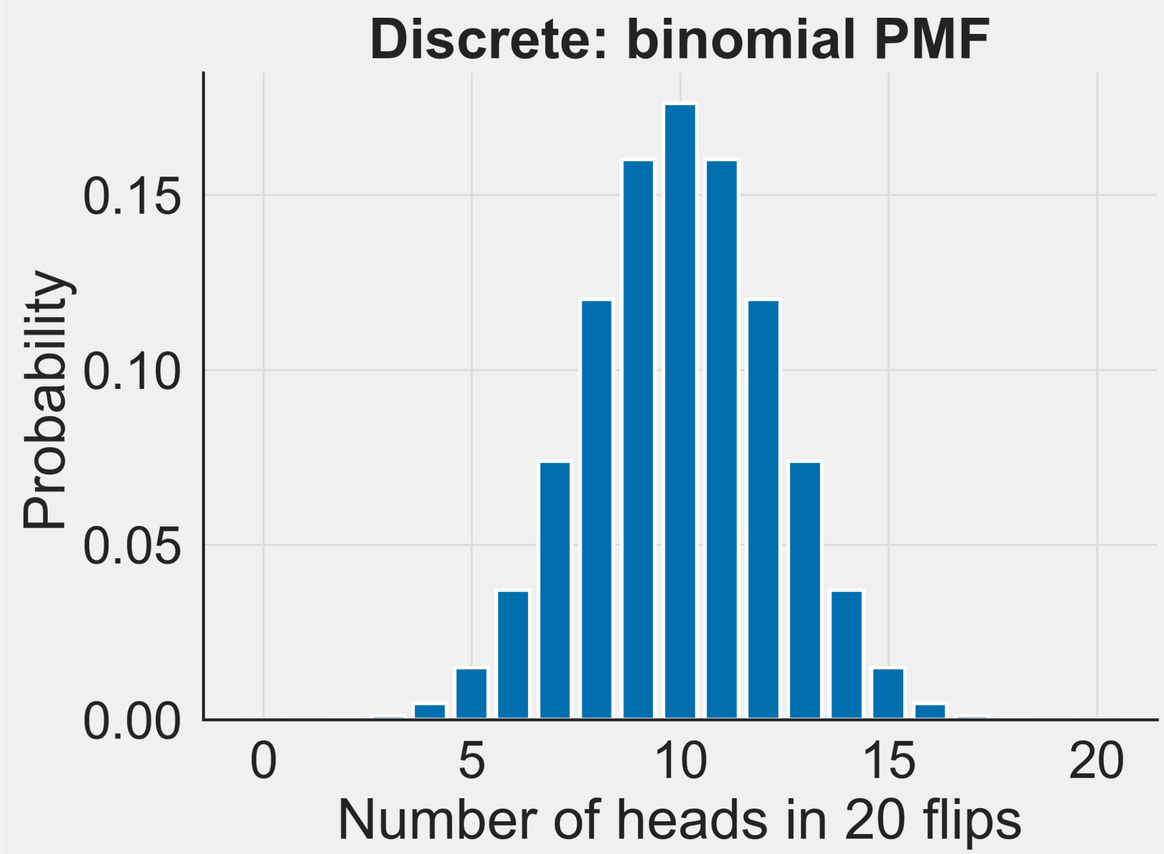
The probabilities sum to 1.

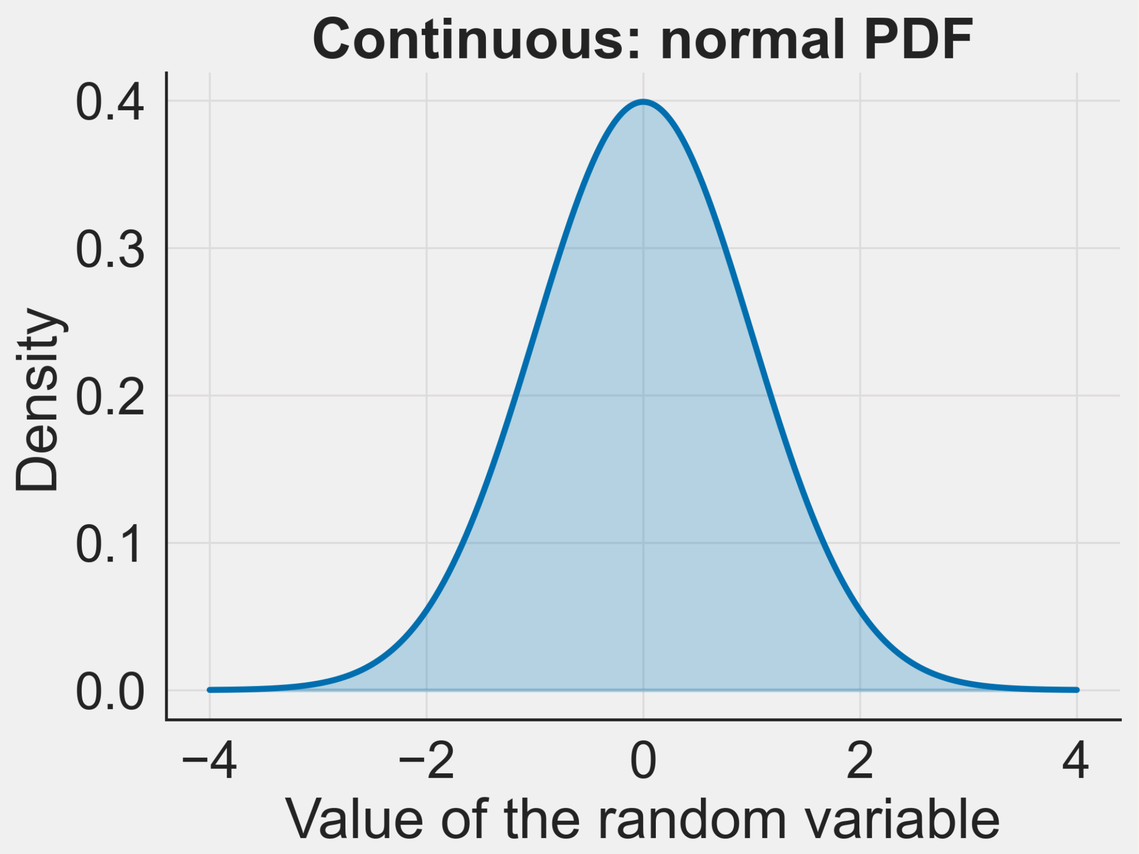
PMF and PDF

- **Probability mass function (PMF)**: for a discrete random variable, the probability that it equals each value
- **Probability density function (PDF)**: for a continuous random variable, the height of the density curve at each value

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Discrete vs. Continuous





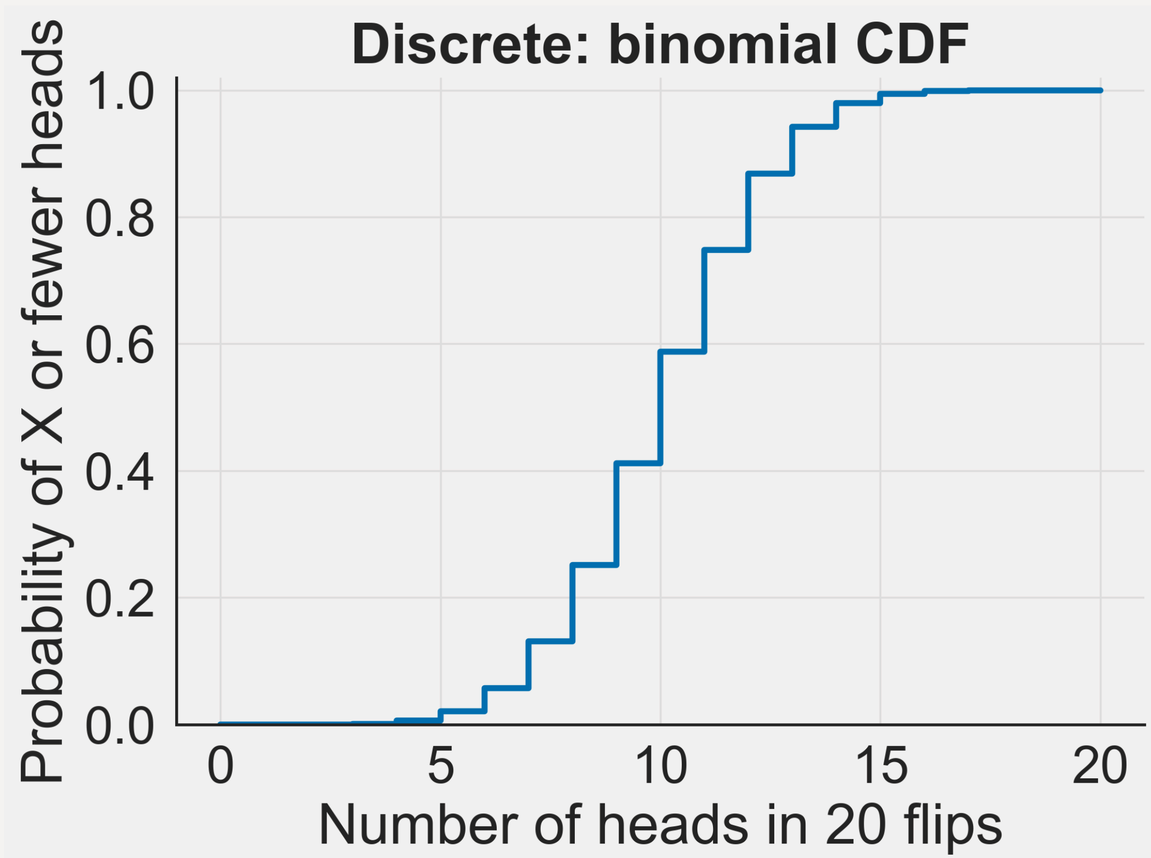
Cumulative Distribution Function

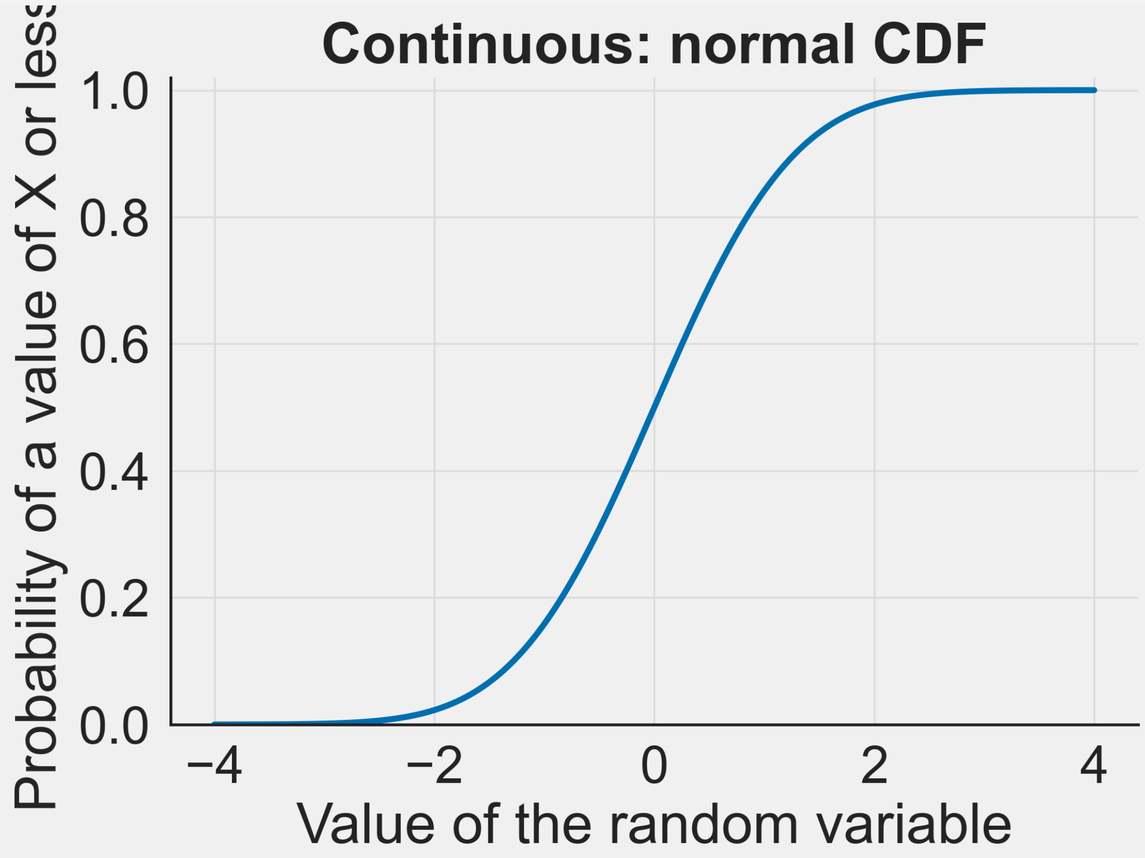
The CDF gives the probability that a random variable is less than or equal to a value:

$$F(x) = P(X \leq x)$$

Discrete: cumulative mass function (CMF). Continuous: CDF.

Reading a CDF





Parameters Define Distributions

- **Normal**: $N(\mu, \sigma^2)$
- **Standard normal**: $Z \sim N(0, 1)$
- **Bernoulli**: $B(1, p)$, one trial that succeeds with probability p
- **Binomial**: $B(n, p)$, the number of successes in n trials
- **Poisson**: $\text{Pois}(\lambda)$, the number of events in an interval at rate λ

Samples and Populations

Parameters and Statistics

Populations have parameters. Samples have statistics.

	Population parameter	Sample statistic
Mean	$\mathbb{E}(X)$	$\bar{X}$
Variance	$\mathbb{V}(X)$	$\hat{\sigma}^2$

Even when everything goes right, a statistic does not equal its parameter.

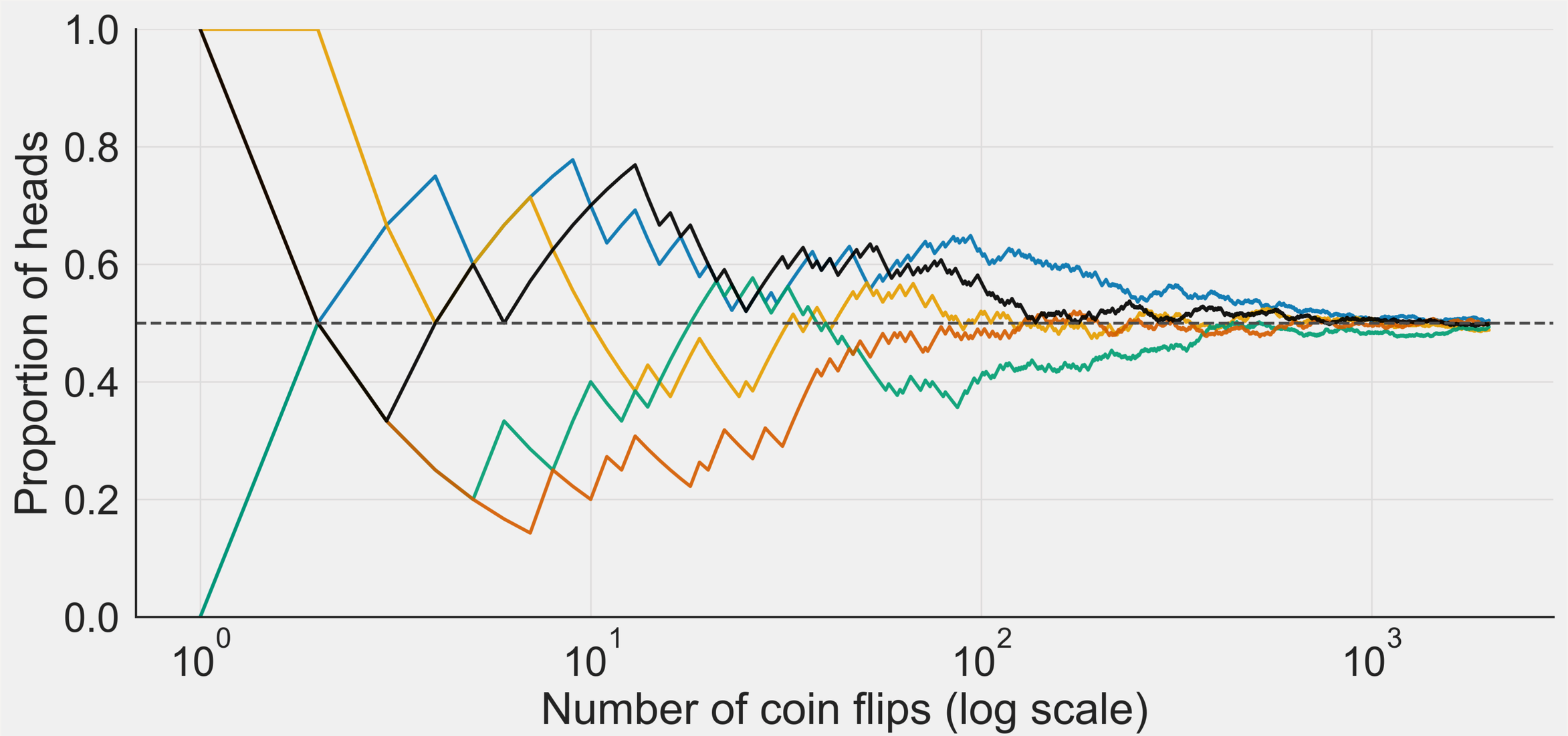
The Law of Large Numbers

For i.i.d. draws, the sample mean converges to the population mean:

$$\lim_{n \to \infty} \bar{X}_n = \mathbb{E}(X)$$

For a binary variable, the sample mean is a proportion and converges to p.

Sample Means Settle Down



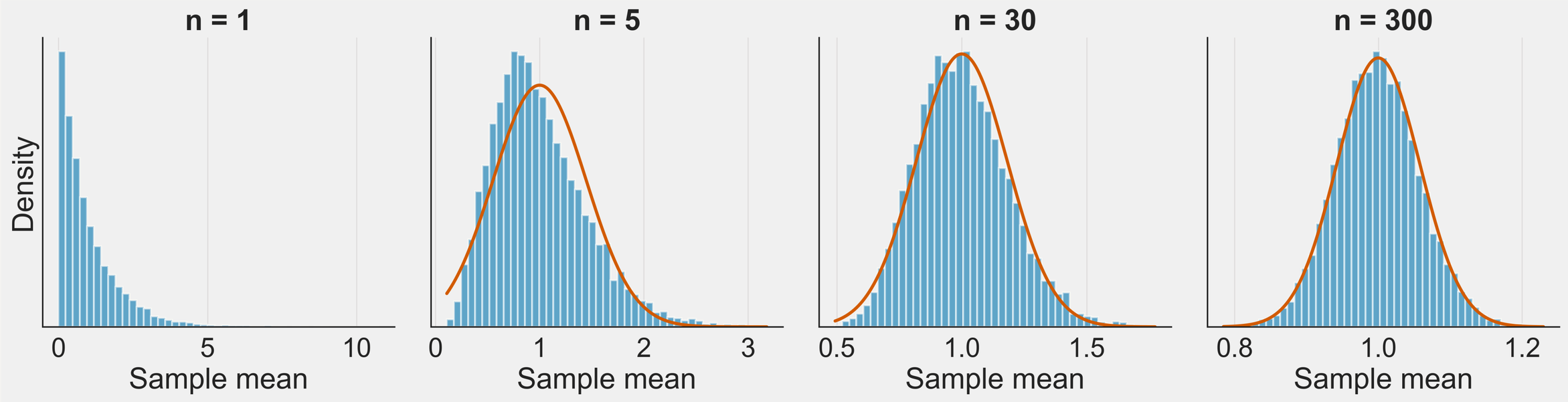
The Central Limit Theorem

As n grows, the standardized sample mean is approximately standard normal:

$$\frac{\bar{X} - \mathbb{E}(X)}{\sqrt{\mathbb{V}(X)/n}} \sim N(0,1)$$

This holds even when X itself is not normal.

Skewed Data, Normal Means



Why We Care About the Normal

- Hypothesis tests are about population parameters
- We estimate parameters with sample means or regression coefficients
- By the CLT, these estimators are approximately normal in large samples
- So the normal distribution is the reference distribution for most tests

What Is Your Estimand?

Start With the Question

Much of social science is not even wrong. It is not specific enough to be wrong.

What is the estimand?

Hypothesizing in the Model

Much of social science reduces to a hypothesis about a regression coefficient.

This defines the research goal inside a statistical model. The results:

- HARKing (hypothesizing after the results are known)
- Bad controls
- Unstated assumptions and uninformative models
- Undefined research goals

Theoretical Estimand

Two pieces:

- **Unit-specific quantity**: what we would like to know about each unit
- **Target population**: the units we want to aggregate across

Together: what we would know if we had data on the full population, in all factual and counterfactual scenarios.

Empirical Estimand

An empirical estimand is an observable quantity linked to the theoretical estimand by assumptions.

Example: a difference in mean test scores between students randomly assigned to small and regular classes.

Estimation Strategy

An estimator is a rule for computing an estimate from data.

Example: the difference-in-means estimator, or OLS.

From Theory to Data

1. Theoretical estimand, linked by argument
2. Empirical estimand, linked by assumption
3. Estimation strategy, linked by data

Why State the Estimand First?

- It forces a clear claim about the world
- It forces the theorized relationship to be explicit
- It makes the assumptions visible
- It requires you to justify the estimator

Vocabulary

- **Parameter**: a true, fixed characteristic of a population
- **Estimand**: the specific quantity we want to estimate, often a parameter, but possibly a causal effect such as the ATE
- **Estimator**: the rule used to compute an estimate from data
- **Estimate**: the number we get when we apply an estimator to a sample

Uncertainty

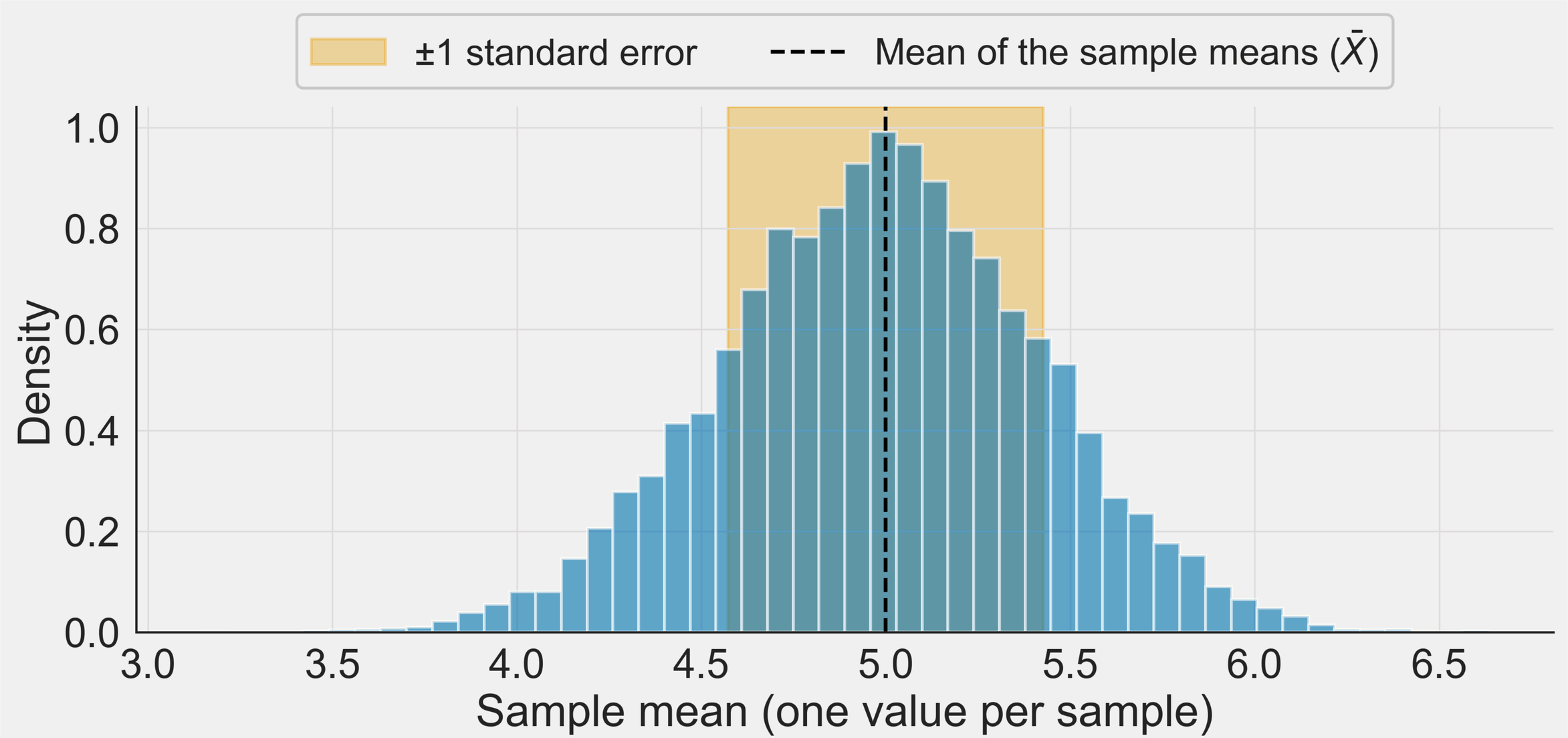
The Sampling Distribution

The sampling distribution describes how an estimator varies from sample to sample.

For the estimators in this course, it is approximately normal and centered on the parameter:

$$\hat{\theta} \sim N\left(\theta, \mathrm{SE}^2\right)$$

Sampling Distribution of a Mean



Standard Error

The standard error is the standard deviation of the sampling distribution.

For the sample mean:

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

In practice we estimate σ with the sample standard deviation $\hat{\sigma}$.

Estimating the SE: Two Variance Rules

For a constant a and independent random variables X_1 and X_2:

$$\mathrm{Var}(aX) = a^2 \, \mathrm{Var}(X)$$

$$\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2)$$

Estimating the SE: Variance of the Sample Mean

Take n independent draws $X_1, \ldots, X_n$, each with variance σ^2:

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n} \sum_{i=1}^{n} X_i \right)$$

Pull out the constant $\frac{1}{n}$, which becomes $\frac{1}{n^2}$:

$$= \frac{1}{n^2} \mathrm{Var}\left(\sum_{i=1}^{n} X_i \right)$$

Estimating the SE: Variance of the Sample Mean

The draws are independent, so the variances add:

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$$

The standard error is the square root of this variance:

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Standard Errors for Regression

For OLS, the standard errors come from the variance of $\widehat{\boldsymbol{\beta}}$:

$$\widehat{\mathrm{Var}}(\widehat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\mathbf{X}^\top\mathbf{X})^{-1}$$

The standard error of $\widehat{\beta}_j$ is the square root of the jth diagonal element.

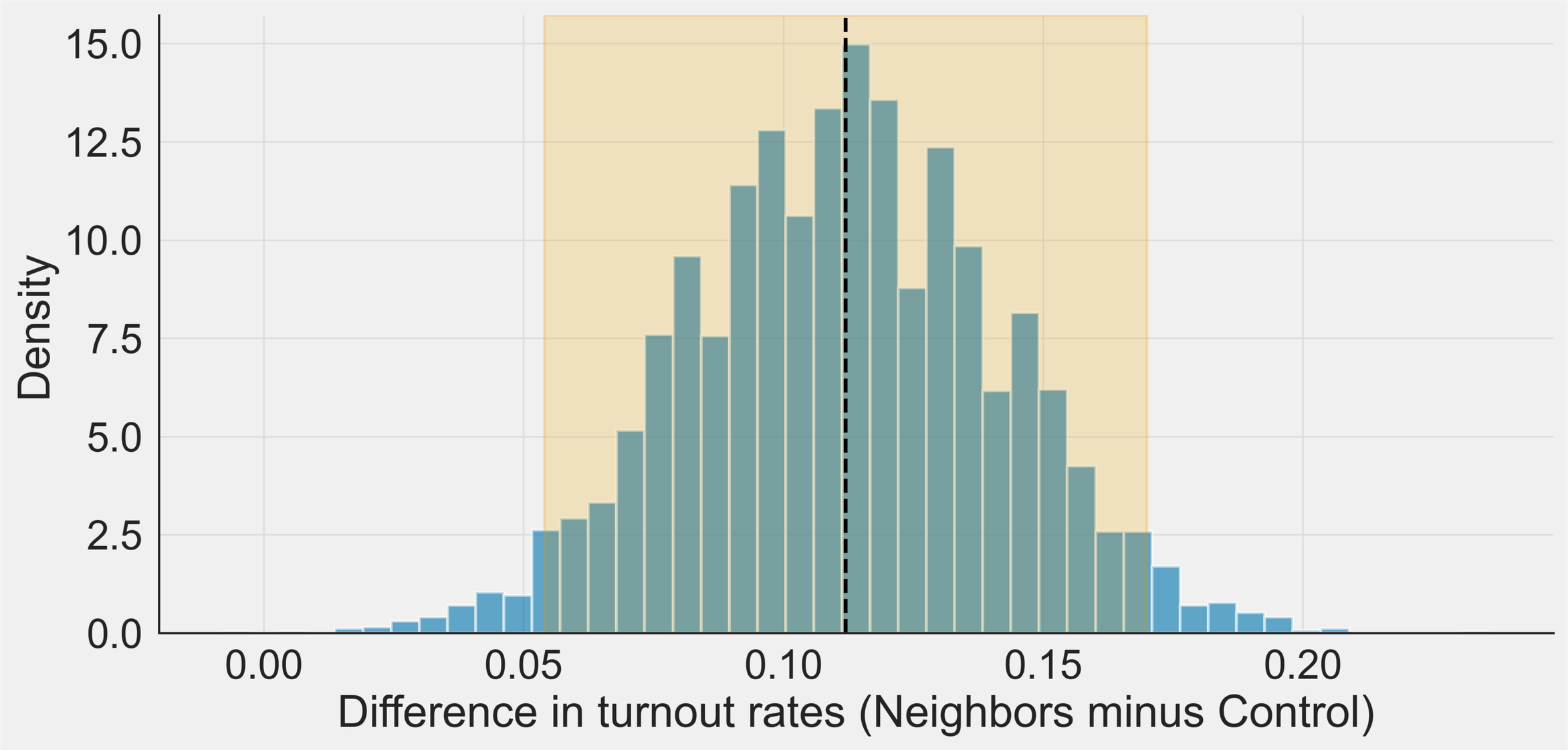
We estimate σ^2 from the residuals:

$$\hat{\sigma}^2 = \frac{\widehat{\boldsymbol{\varepsilon}}^\top\widehat{\boldsymbol{\varepsilon}}}{n-k}$$

Bootstrapped Standard Errors

1. Define the statistic and collect the data
2. Resample the data with replacement, same n
3. Compute the statistic on each resample
4. Take the standard deviation of the computed statistics

Bootstrap Distribution



Z-Scores

A z-score is how many standard deviations a value is from the mean:

$$Z = \frac{X - \mu}{\sigma}$$

If $X \sim N(\mu, \sigma^2)$, then $Z \sim N(0, 1)$.

A 1 unit change in Z is a 1 standard deviation change. About 95% of the standard normal lies within 1.96 of zero.

The Standardized Estimator

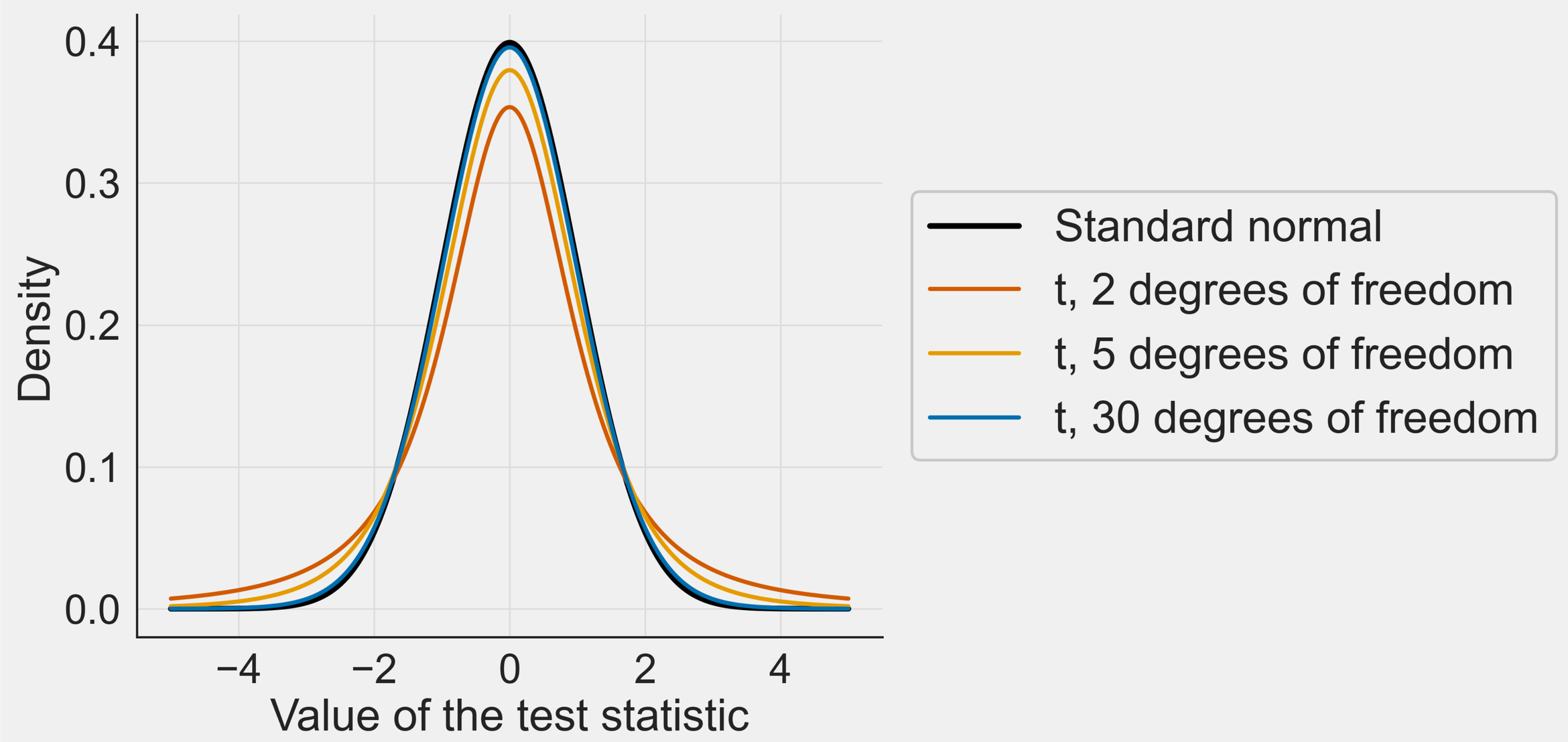
Subtract the true value and divide by the standard error:

$$\frac{\hat{\theta} - \theta}{\text{SE}} \sim N(0, 1)$$

The t-Distribution

- A z-score assumes the standard error is known, or that n is large
- We estimate the standard error, which adds uncertainty
- The t-distribution accounts for this, with $n - k$ degrees of freedom
- As the degrees of freedom grow, t approaches the normal distribution

The t-Distribution



The t-Statistic

Replace the unknown standard error with an estimate:

$$t = \frac{\hat{\theta} - \theta}{\widehat{\mathrm{SE}}} \sim t_{n-k}$$

Confidence Intervals

A confidence interval is a range of values that is likely to include the true value of the parameter.

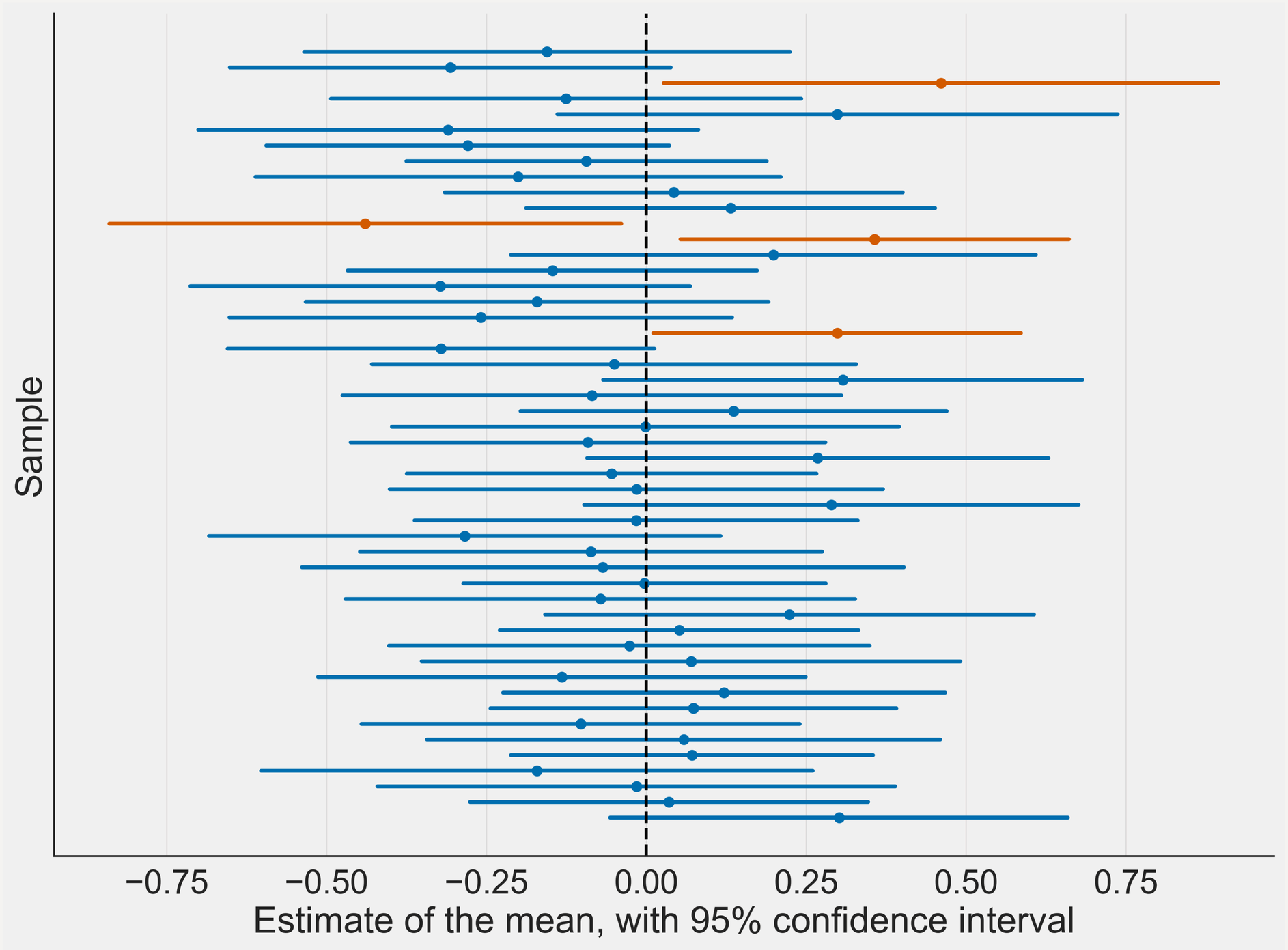
$$\hat{\theta} \pm 1.96 \times \widehat{\mathrm{SE}}$$

Interpreting a Confidence Interval

If we drew many samples and built a 95% interval from each, about 95% of the intervals would contain the true value.

It is not the probability that the true value is in this interval.

Confidence Intervals Across Samples



Hypothesis Testing

Key Terms

- **Statistic**: a quantity computed from a sample
- **Test statistic**: a statistic used to test a null hypothesis, usually a z or t statistic
- **Null hypothesis** (H_0): the claim we test, usually that the parameter equals a specific value
- **p-value**: the probability of a test statistic at least as extreme as the one observed, assuming the null is true
- **Significance level** (α): the cutoff for rejecting the null

The NHST Procedure

1. State a null hypothesis and an alternative
2. Choose a test statistic and a significance level α
3. Derive the reference distribution: the distribution of the test statistic if the null is true
4. Compute the p-value
5. Reject the null if the p-value is less than or equal to α

The Wald Test

Test whether a parameter equals a specific value:

$$H_0 : \theta = \theta_0 \qquad H_1 : \theta \neq \theta_0$$

$$t = \frac{\hat{\theta} - \theta_0}{\widehat{\mathrm{SE}}}$$

The statistic is the distance from the estimate to the null value, in standard errors.

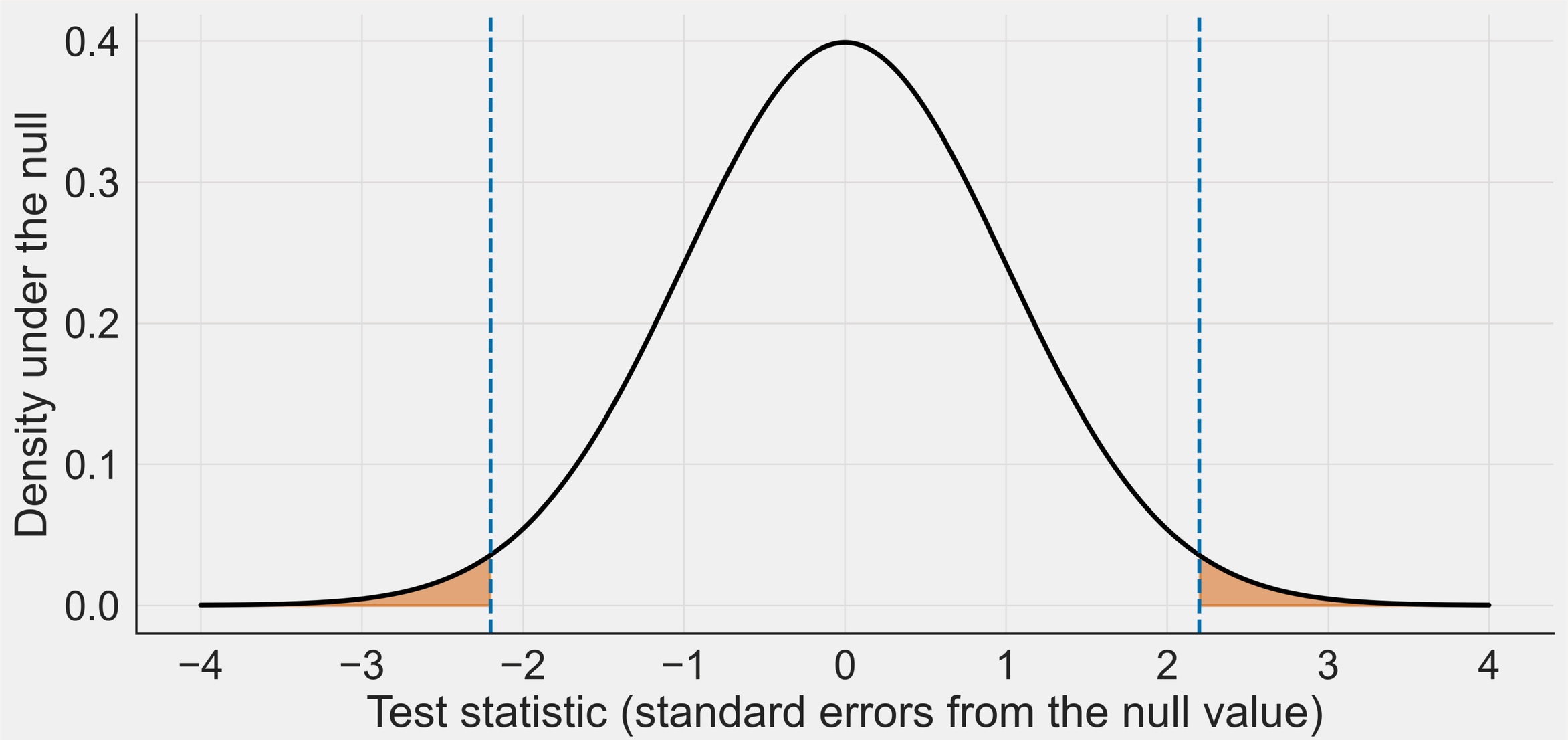
Testing Against Zero

In a regression, the default null is that a coefficient is zero:

$$H_0 : \beta_j = 0 \qquad H_1 : \beta_j \neq 0$$

$$t = \frac{\hat{\beta}_j}{\widehat{\mathrm{SE}}(\hat{\beta}_j)}$$

The p-Value


0.4
0.3
0.2
0.1
0.0
Density under the null
−4
−3
−2
−1
0
1
2
3
4
Test statistic (standard errors from the null value)

One-Sided vs. Two-Sided Tests

- **Two-sided**: detects a deviation from the null in either direction
- **One-sided**: detects a deviation in one direction only

Use a one-sided test only when theory predicts a direction and evidence in the other direction is irrelevant.

Difference in Means

Do Neighbors mailers raise turnout relative to the control group?

	Turnout rate
Neighbors (n = 500)	0.398
Control (n = 500)	0.286
Difference	0.112

$$t = \frac{0.112 - 0}{0.030} \approx 3.76 \qquad p < 0.001$$

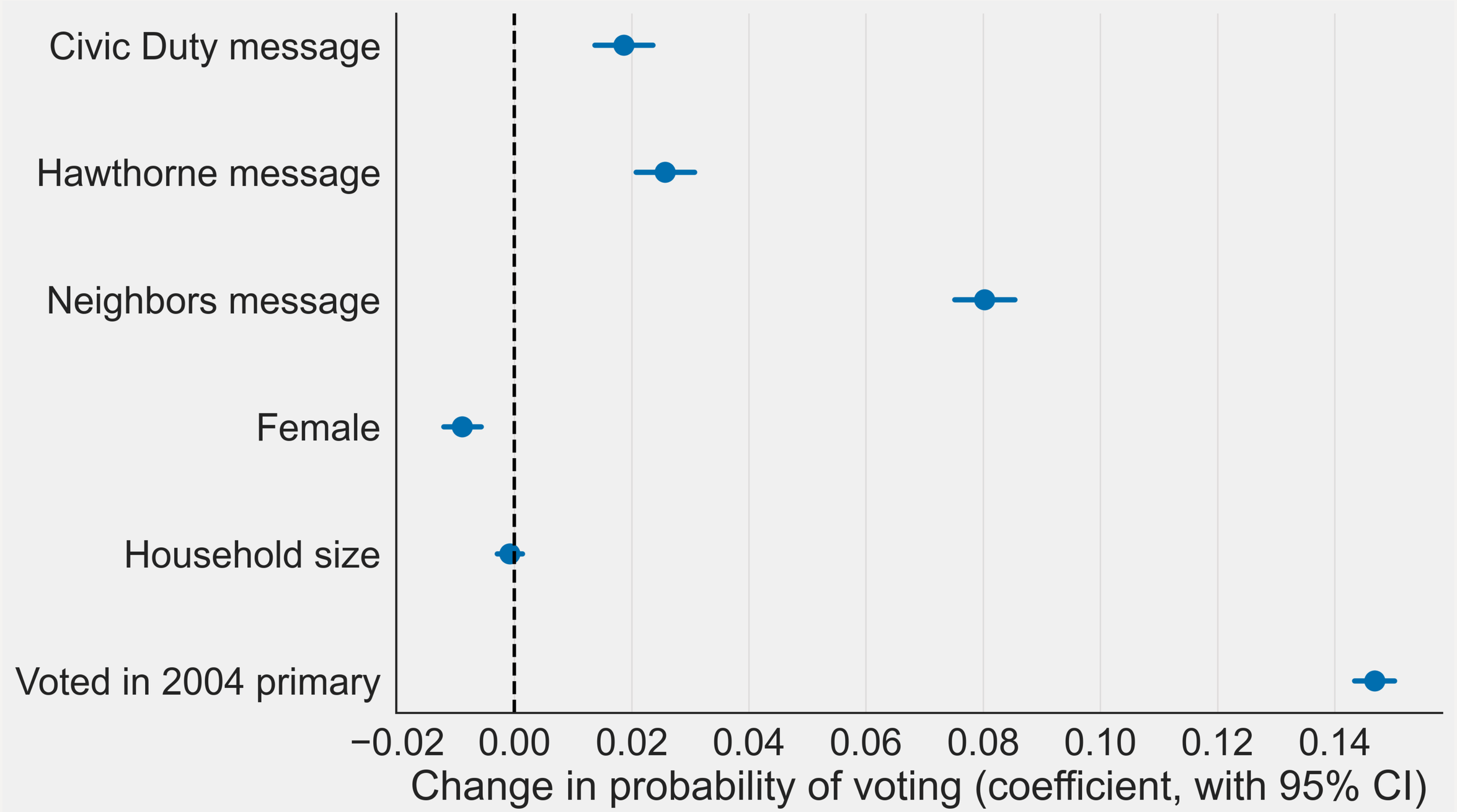
Reading a Regression Table

Outcome: voted in the 2006 primary

	(1)	(2)
Civic Duty message	0.018***	0.019***
	(0.003)	(0.003)
Hawthorne message	0.026***	0.026***
	(0.003)	(0.003)
Neighbors message	0.081***	0.080***
	(0.003)	(0.003)
Female		−0.009***
		(0.002)
Age (decades)		0.039***
		(0.001)
Household size		−0.001
		(0.001)
Voted in 2004 primary		0.147***
		(0.002)
Constant	0.297***	0.051***
	(0.001)	(0.005)
Observations	305,866	305,866
R^2	0.003	0.044

Standard errors in parentheses. The control group is the omitted category. *p < 0.05, **p < 0.01, ***p < 0.001

Forest Plots

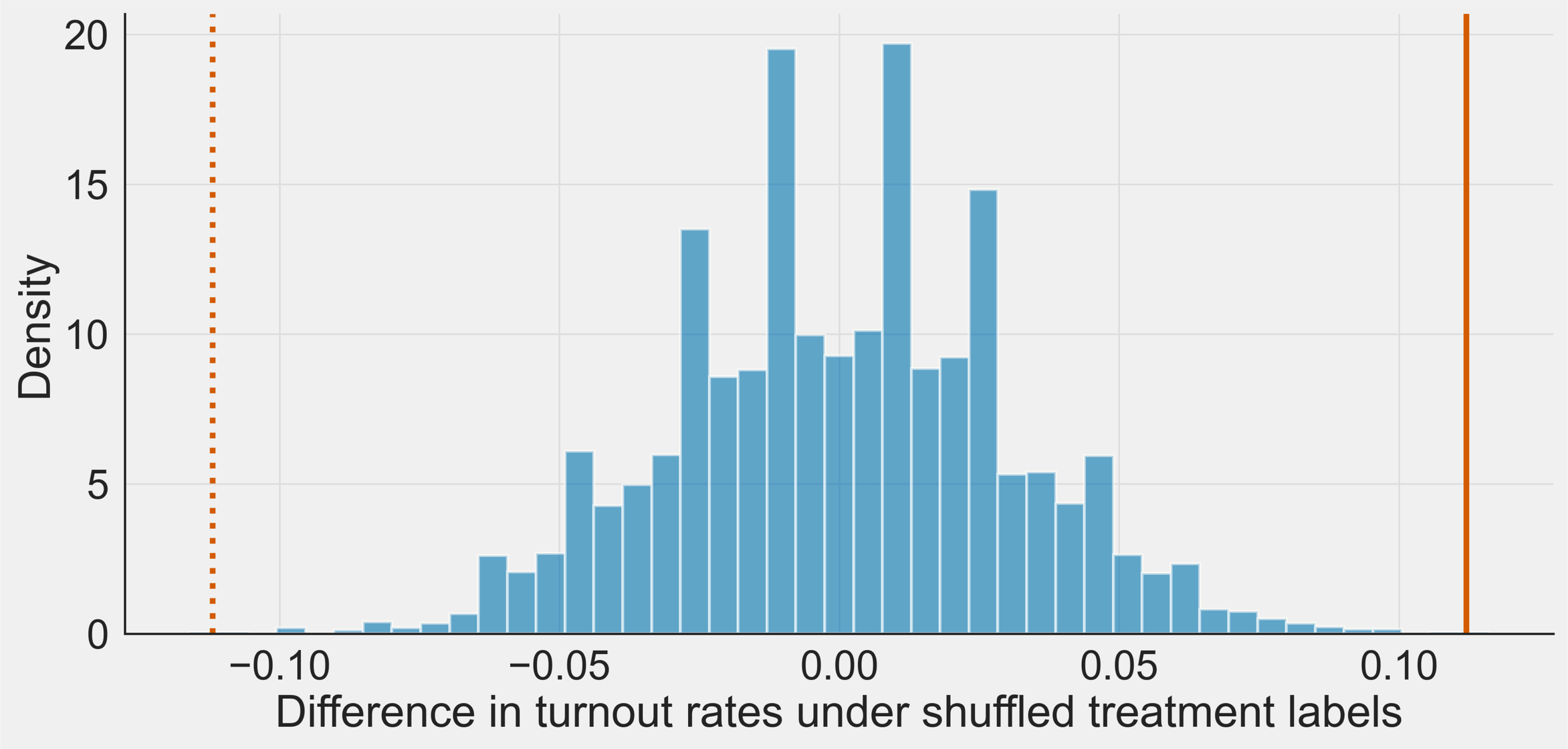


Permutation Tests

A permutation test builds the null distribution from the data instead of assuming one.

1. Compute the observed test statistic
2. Randomly shuffle the treatment labels
3. Recompute the statistic on the shuffled data
4. Repeat many times to build the null distribution
5. The p-value is the share of shuffled statistics at least as extreme as the observed one

Permutation Null Distribution



Chi-Squared Tests

When to Use a Chi-Squared Test

	t-test or Wald test	Chi-squared test
Null hypothesis	One parameter equals a value	Counts follow a specified pattern
Variables	Numeric outcome	Categorical variables
Statistic	Distance from the null in standard errors	Gap between observed and expected counts
Tells you	Direction and size of an effect	Whether the pattern differs from the null

With two categories on each variable, the two tests agree.

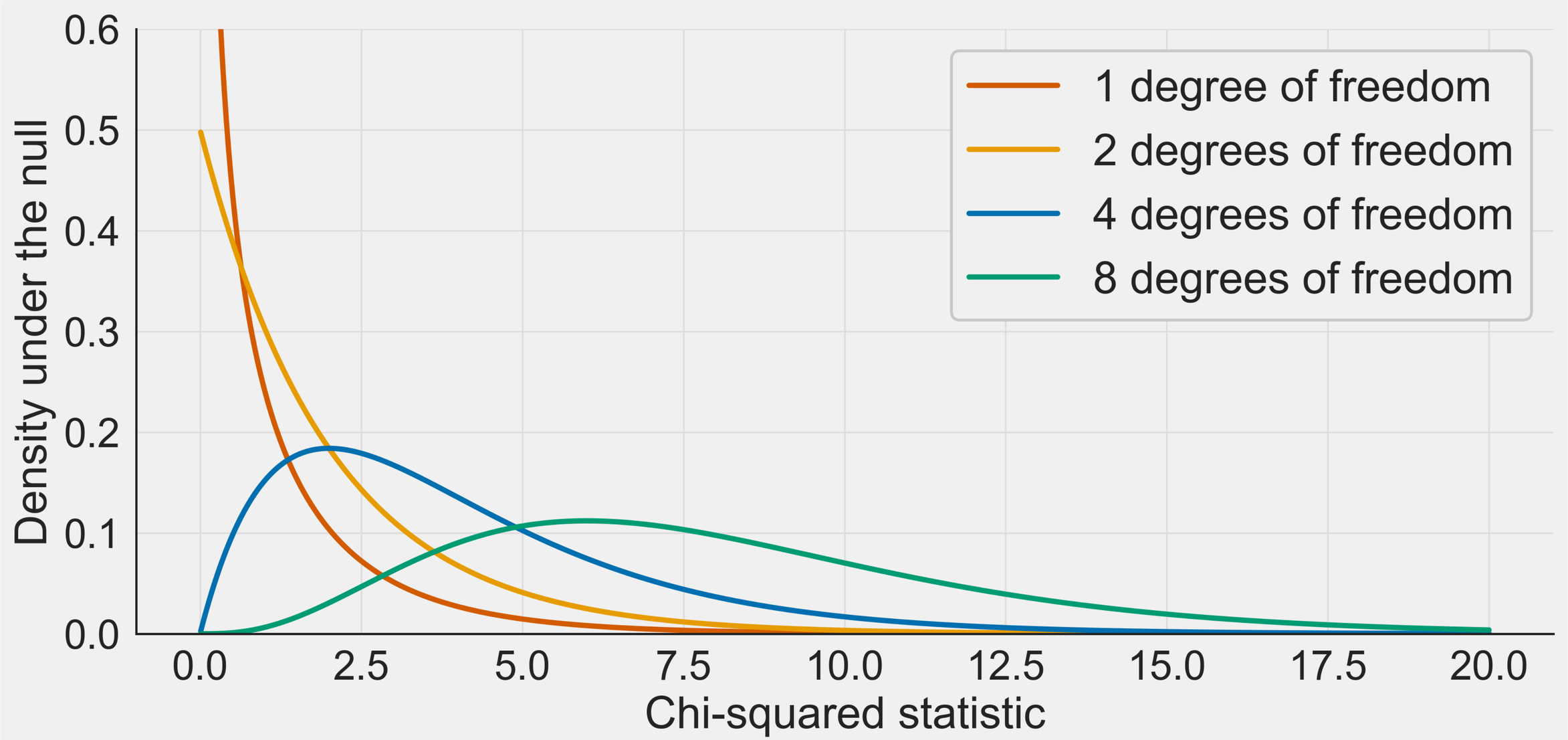
The Chi-Squared Distribution

Let $Z_1, \ldots, Z_d$ be independent standard normal random variables. The sum of their squares is a chi-squared random variable with d degrees of freedom:

$$\chi^2_d = Z_1^2 + Z_2^2 + \cdots + Z_d^2$$

- It has one parameter, the degrees of freedom d
- It takes only positive values and is right-skewed
- Its mean is d

Shape of the Chi-Squared



Tests With Counts

Is turnout related to which mailer a household received?

- Both variables are categorical, so we have counts in each cell of a table
- **Null hypothesis**: the two variables are independent
- **Test**: compare the observed counts to the counts expected if the null is true

Observed Counts

	Did not vote	Voted	Total	Turnout
Control	134,513	56,730	191,243	29.7%
Civic Duty	26,197	12,021	38,218	31.5%
Hawthorne	25,888	12,316	38,204	32.2%
Neighbors	23,763	14,438	38,201	37.8%
Total	210,361	95,505	305,866	31.2%

Expected Counts

If the variables are independent, the expected count for a cell is:

$$E = \frac{(\text{row total}) \times (\text{column total})}{n}$$

Control, voted: $E = \dfrac{191{,}243 \times 95{,}505}{305{,}866} \approx 59{,}715$

The observed count is 56,730.

The Chi-Squared Statistic

$$\chi^2 = \sum_{\text{cells}} \frac{(O - E)^2}{E}$$

- O is the observed count and E is the expected count
- Squaring the gap means over- and under-counts both add to the statistic
- Dividing by E puts each gap in units of its own noise

Degrees of Freedom

For a table with r rows and c columns:

$$df = (r-1)(c-1)$$

Here $r = 4$ and $c = 2$, so $df = 3 \times 1 = 3$.

Computing the Test

$$\chi^2 = 1004.0 \qquad df = 3$$

The 5% critical value is 7.81. The p-value is about 10^{-217}, reported as $p < 0.001$.

We reject independence: turnout is related to the mailer.

The Neighbors cells contribute 76% of the statistic.

Goodness-of-Fit Test

Compare the counts of one categorical variable to hypothesized proportions.

$$\chi^2 = \sum_{\text{categories}} \frac{(O-E)^2}{E} \qquad df = (\text{categories}) - 1$$

Example: were households assigned to the four groups in the designed proportions?

$\chi^2 = 0.09, df = 3, p = 0.99$

Conditions and Limits

- Expected counts should be at least 5 in nearly every cell
- Observations must be independent
- The test uses counts, not percentages
- It detects an association, and does not measure its strength
- With large n, a trivial association is significant

Fit-Based Tests

A Naive Model

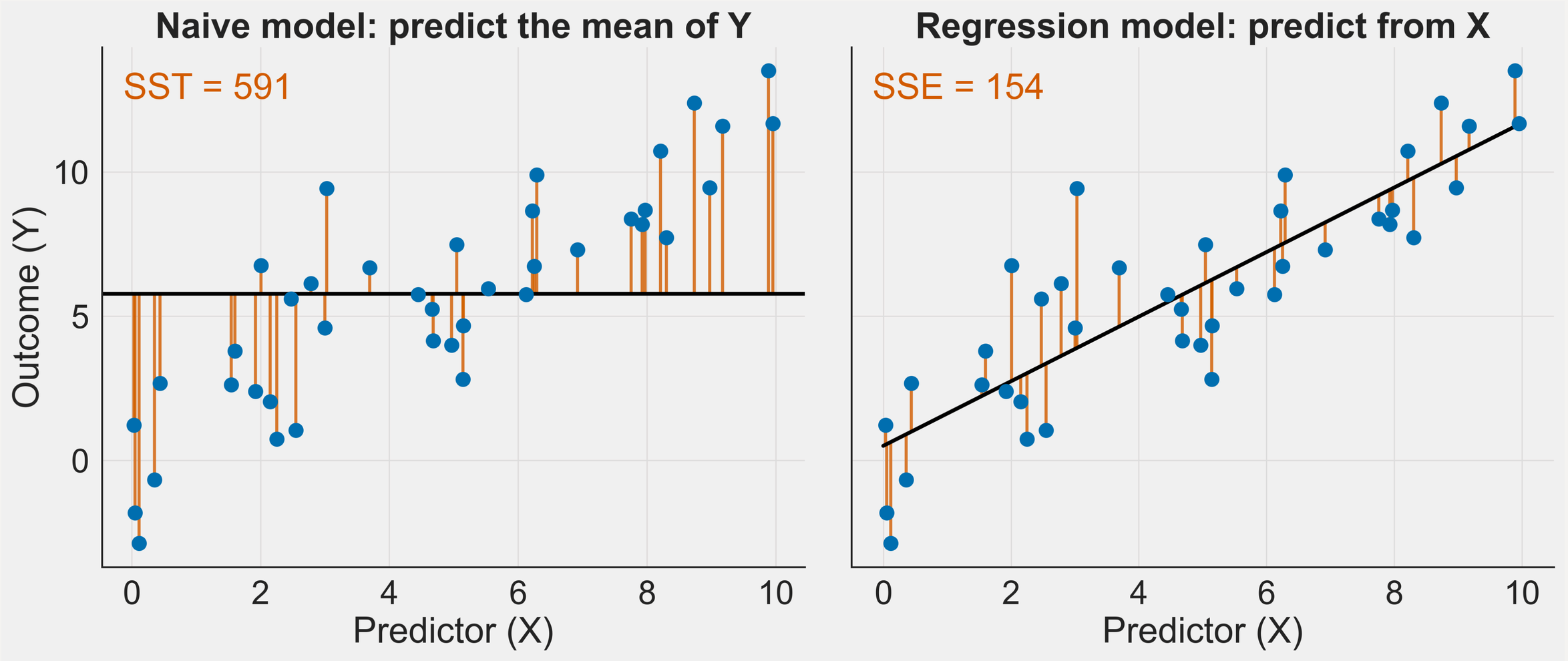
Suppose we ignore X and predict the sample mean $\bar{Y}$ for every observation.

Its residuals are $Y_i - \bar{Y}$. Squaring and summing them gives the total sum of squares:

$$\text{SST} = \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

SST is the SSE of the naive model.

Naive vs. Regression Residuals



R-Squared

How much of the naive model's error is left after we use the model?

$$\frac{\text{SSE}}{\text{SST}}$$

The share that is removed:

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}$$

R^2 is the share of the variance in Y explained by the model.

Reading R-Squared

- $R^2 = 0$: SSE equals SST, and the model predicts no better than $\bar{Y}$
- $R^2 = 1$: SSE is zero, and the model predicts every observation exactly
- $R^2 = 0.74$: the model removes 74% of the naive model's squared error
- Turnout model: $R^2 = 0.044$

R-Squared and Correlation

With one predictor, R^2 is the squared correlation between X and Y:

$$R^2 = r_{XY}^2$$

With several predictors, it is the squared correlation between Y and the fitted values:

$$R^2 = r_{Y\widehat{Y}}^2$$

r_{XY} is the sample analogue of the population correlation ρ_{XY} .

Why R^2 Equals r^2

In simple regression, $\widehat{Y}_i - \bar{Y} = \hat{\beta}(X_i - \bar{X})$, so the variation the model explains is:

$$\text{SST} - \text{SSE} = \sum_{i=1}^{n}(\widehat{Y}_i - \bar{Y})^2 = \hat{\beta}^2 \sum_{i=1}^{n}(X_i - \bar{X})^2$$

Substitute $\hat{\beta} = r_{XY} \dfrac{s_Y}{s_X}$ and $\sum_{i=1}^{n}(X_i - \bar{X})^2 = (n-1)s_X^2$:

$$\text{SST} - \text{SSE} = r_{XY}^2 \dfrac{s_Y^2}{s_X^2}(n-1)s_X^2 = r_{XY}^2 (n-1)s_Y^2 = r_{XY}^2 \text{ SST}$$

From Explained Variation to R^2

The last slide showed that the explained variation is r_{XY}^2 times SST. Divide both sides by SST:

$$R^2 = \frac{\text{SST} - \text{SSE}}{\text{SST}} = r_{XY}^2$$

What R^2 Is Good For

- Comparing models of the same outcome, using adjusted R^2 when they differ in size
- Flagging an underspecified model, when a low R^2 says much of Y is unexplained
- Benchmarking against a theoretical maximum, such as the reliability of the outcome measure
- Describing how much of the variation a theory accounts for
- Evaluating prediction, when R^2 is computed on held-out data

What R^2 Does Not Tell You

- It never decreases when you add a variable
- A high R^2 does not mean a causal effect was identified
- A low R^2 does not mean the effect is unimportant
- Adjusted R^2 penalizes extra predictors

Restricted and Unrestricted Models

- **Unrestricted model**: includes every predictor under consideration
- **Restricted model**: the same model with q restrictions imposed, usually dropping q variables

The null hypothesis is that the restrictions are true.

The restricted model can never fit better:

$$\text{SSE}_R \geq \text{SSE}_U$$

The F Statistic

$$F = \frac{(\text{SSE}_R - \text{SSE}_U)/q}{\text{SSE}_U/(n-k)}$$

- **Numerator**: the increase in squared error from imposing the restrictions, per restriction
- **Denominator**: the unrestricted model's estimate of the error variance, $\hat{\sigma}^2$

Reading the F Statistic

- If the restrictions are true, the numerator and denominator both estimate σ^2, so $F \approx 1$
- If the restrictions are false, the numerator is larger than σ^2, so $F > 1$
- A large F is evidence against the null

The F Statistic Using R^2

When both models have the same outcome, we can write F with R^2:

$$F = \frac{(R^2_U - R^2_R)/q}{(1 - R^2_U)/(n - k)}$$

The improvement in R^2 per restriction, relative to the unexplained share per degree of freedom.

Why the F-Distribution?

If the null is true and the errors are normal:

- The numerator is a sum of q squared standard normal random variables, scaled by σ^2
- The denominator is a sum of $n - k$ squared standard normal random variables, scaled by σ^2
- A sum of d squared independent standard normal random variables follows a chi-squared distribution with d degrees of freedom

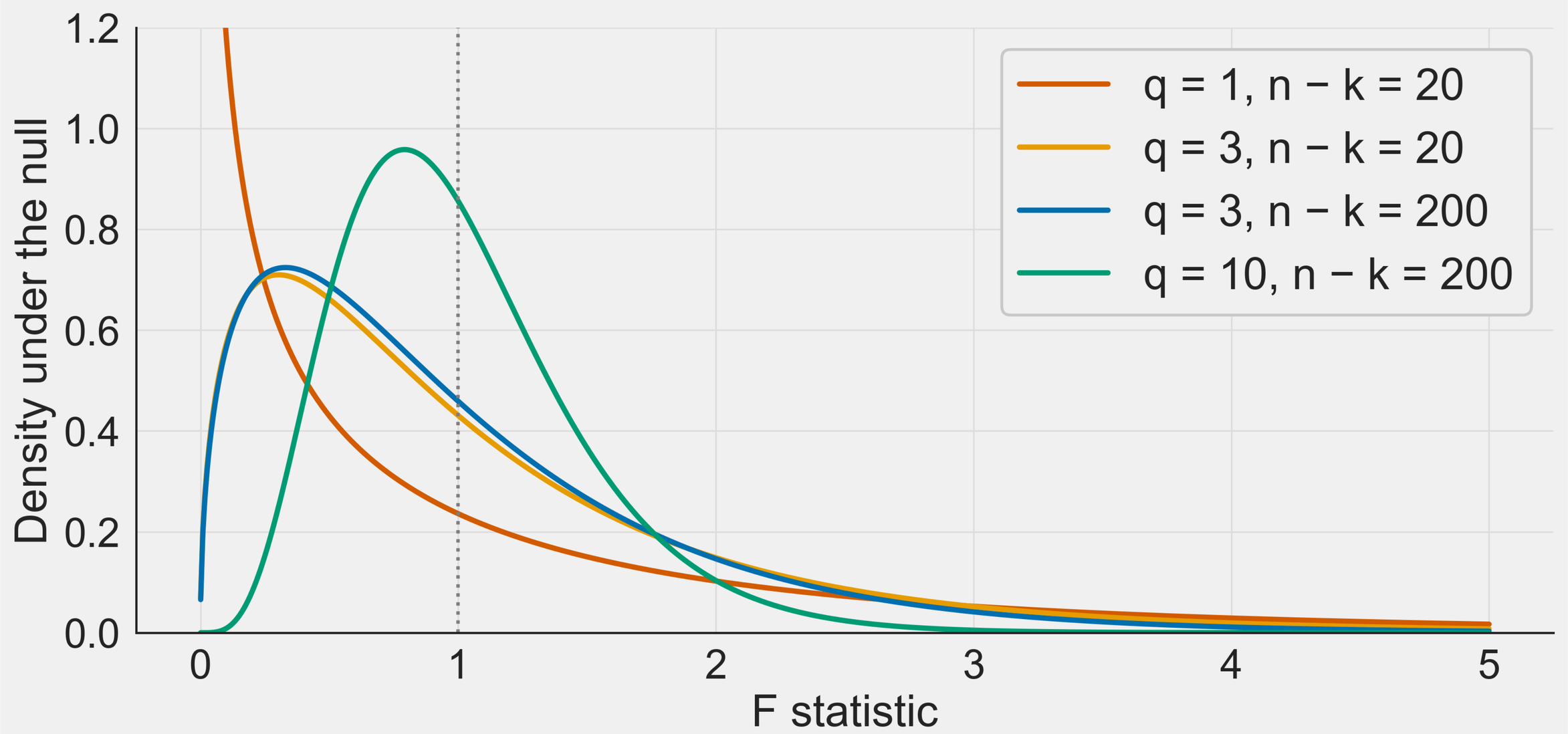
$$F = \frac{\chi^2_q / q}{\chi^2_{n-k} / (n-k)} \sim F_{q,\, n-k}$$

Parameters of the F-Distribution

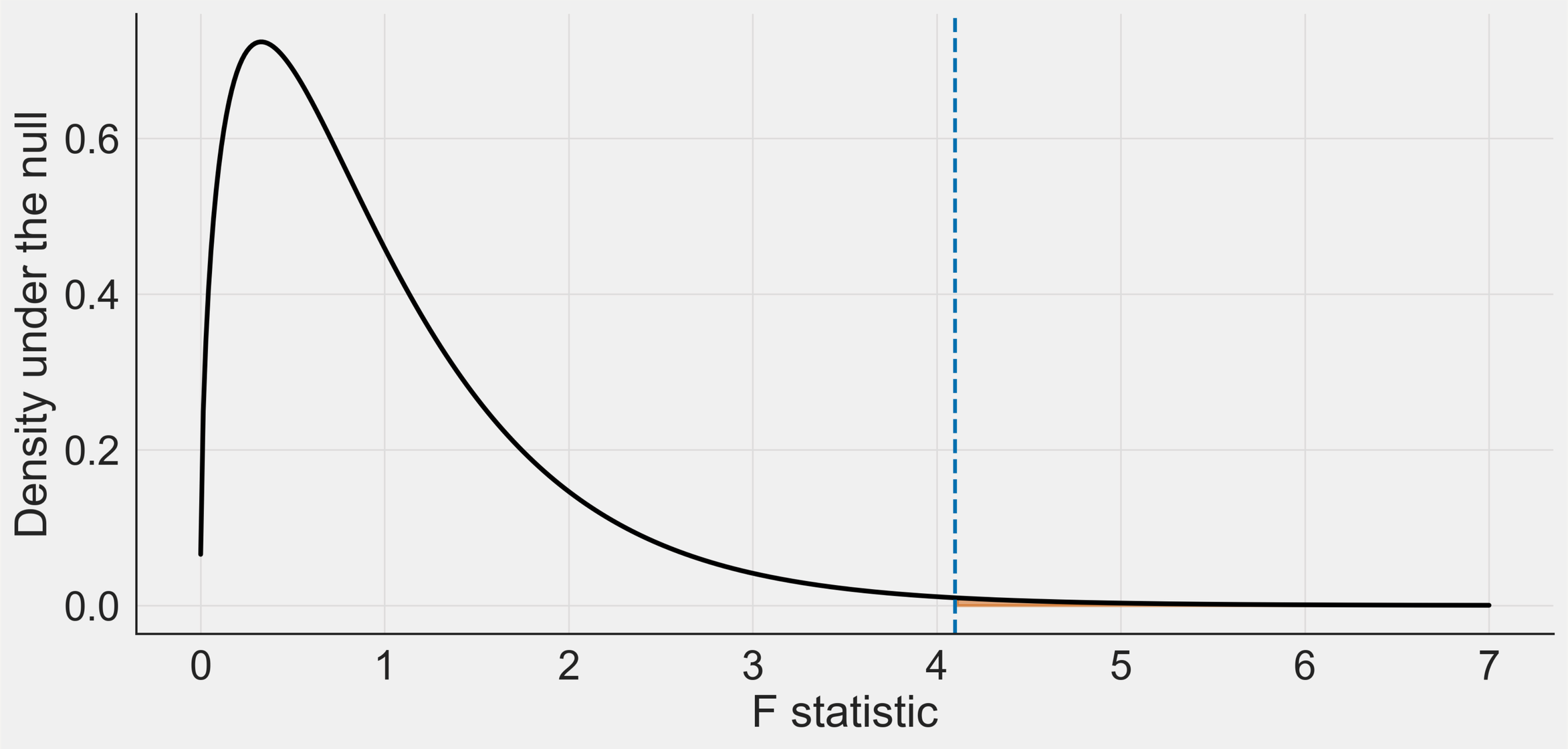
- **Numerator degrees of freedom**: q, the number of restrictions
- **Denominator degrees of freedom**: $n - k$, the observations minus the coefficients in the unrestricted model

We write the distribution as $F_{q,\, n-k}$. It is right-skewed and takes only positive values.

Shape of the F-Distribution



The p-Value for an F-Test



Testing a Group of Coefficients

Do the three mailers jointly change turnout?

$$H_0 : \beta_{\text{Civic Duty}} = \beta_{\text{Hawthorne}} = \beta_{\text{Neighbors}} = 0$$

- Unrestricted model: mailers and covariates, $k = 8$ coefficients
- Restricted model: covariates only, so $q = 3$
- Sample size: $n = 305{,}866$

Computing the F Statistic

$$\text{SSE}_R \approx 62{,}990 \qquad \text{SSE}_U \approx 62{,}780$$

Numerator: $(62{,}990 - 62{,}780)/3 \approx 70.2$

Denominator: $62{,}780/305{,}858 \approx 0.205$

$$F = \frac{70.2}{0.205} \approx 342$$

Computing the p-Value

The reference distribution is $F_{3,\,305858}$.

The 5% critical value is 2.61. Our statistic is 342.

The area to the right of 342 is about 10^{-221}, reported as $p < 0.001$.

We reject the null: the mailers jointly change turnout.

Nested vs. Non-Nested Models

- **Nested**: one model is a special case of the other, obtained by restricting coefficients
- **Non-nested**: neither model is a special case of the other

Nested: covariates only, versus covariates plus mailers.

Non-nested: demographics (female, age, household size), versus history and mailers (voted in 2004, mailers).

The F-test requires nested models.

Information Criteria

AIC and BIC score a model by its fit, plus a penalty for each coefficient. Lower is better.

$$\text{AIC} = n \ln(\text{SSE}/n) + 2k$$

$$\text{BIC} = n \ln(\text{SSE}/n) + k \ln(n)$$

- **Fit**: falls as SSE falls
- **Penalty**: rises with the number of coefficients k

Using AIC and BIC

- Compare models of the same outcome, fit to the same observations
- Lower is better, and only differences matter
- Neither gives a p-value or a hypothesis test
- AIC favors predictive accuracy, BIC favors simpler models

Model	k	R^2	**AIC**	**BIC**
A: demographics	4	0.017	−475,738	−475,696
B: history and mailers	5	0.029	−479,614	−479,561

Out-of-Sample Prediction

1. Split the data into a training set and a test set
2. Fit each model on the training set
3. Predict the outcome in the test set
4. Prefer the model with the smaller prediction error

Prediction error is generally measured with R^2 or RMSE (root mean squared error)

Choosing a Comparison

- Nested models: use an F-test or out-of-sample prediction
- Non-nested models of the same outcome: use AIC, BIC, or out-of-sample prediction
- Better fit does not necessarily mean a better theory

In practice, I am almost always using out-of-sample prediction to compare models.

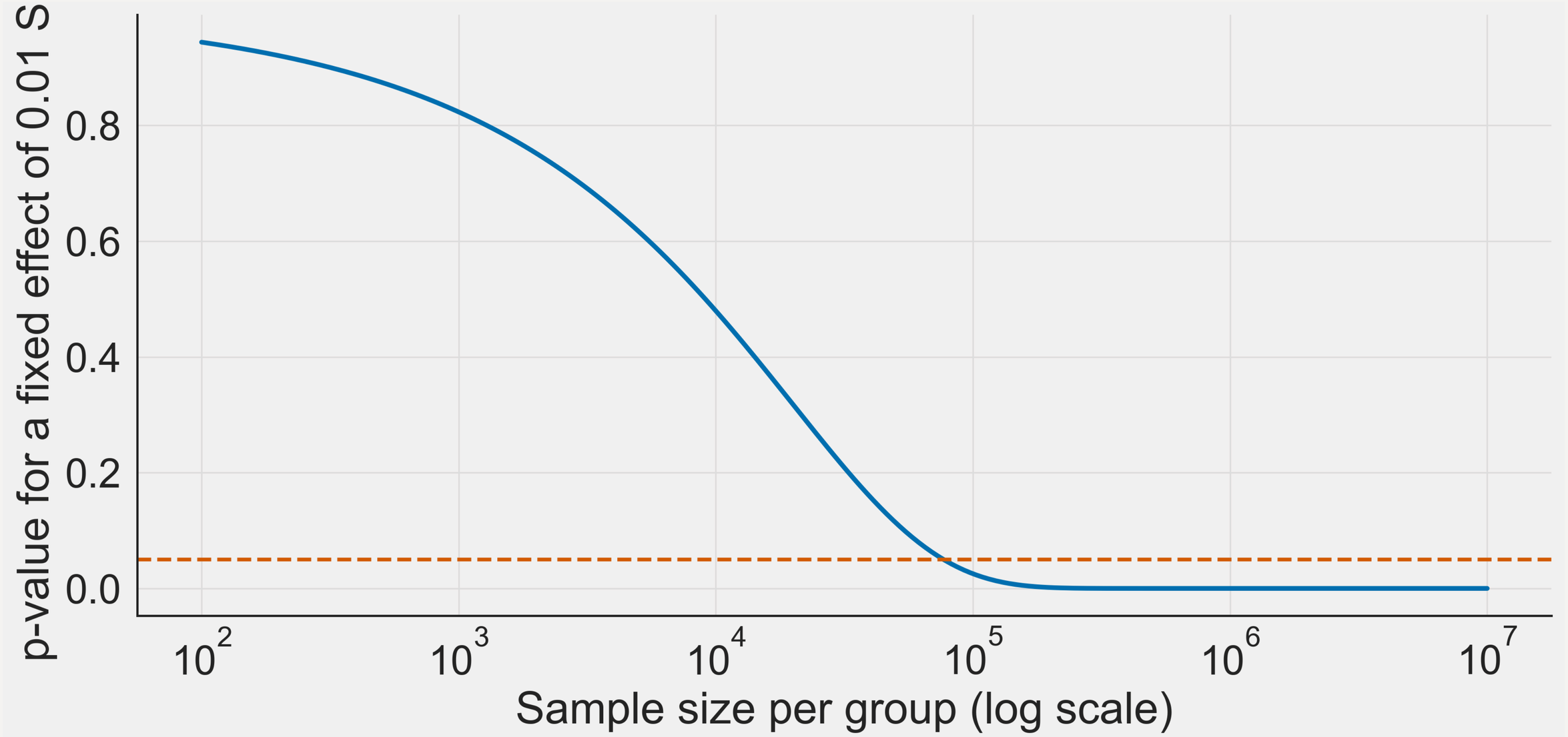
Interpreting Tests Well

Significance vs. Importance

- Statistical significance: is the estimate distinguishable from the null given the sampling noise?
- Practical significance: is the estimate large enough to matter?

With enough data, any nonzero effect becomes statistically significant.

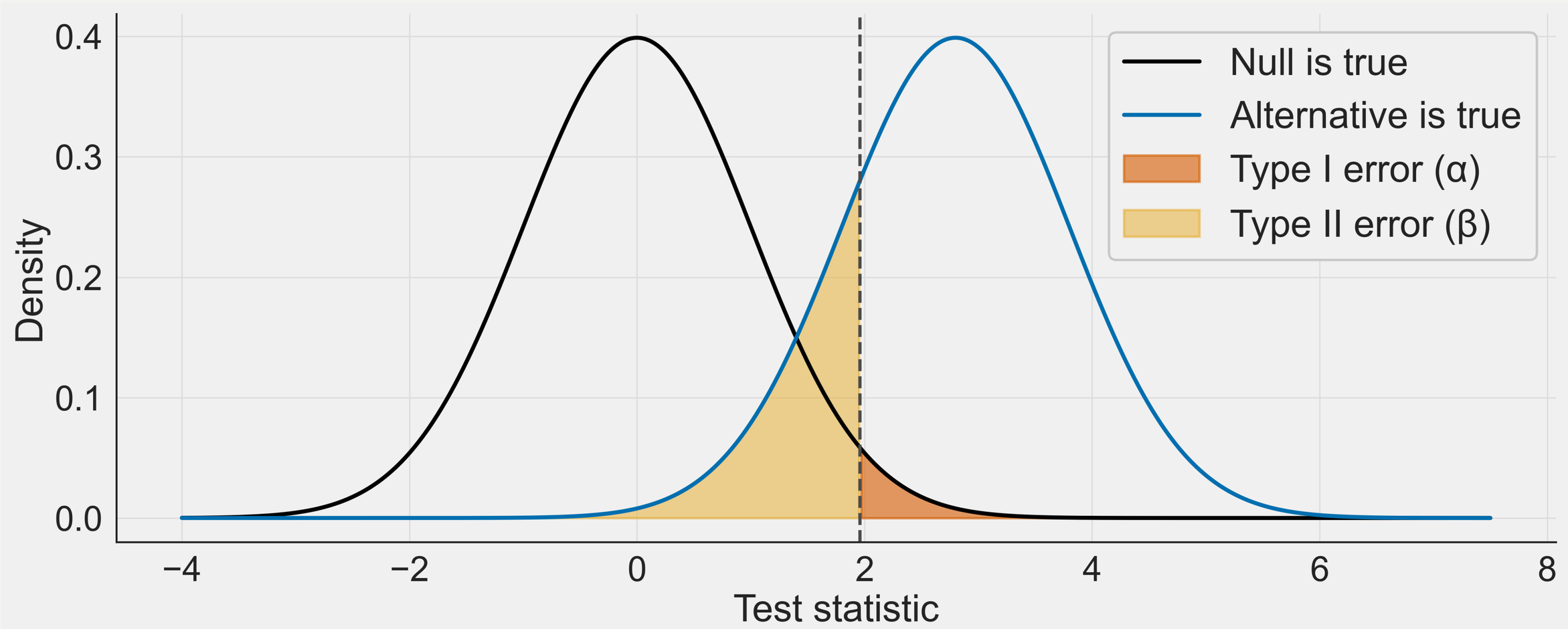
Large Samples, Small Effects



Type I and Type II Errors

- **Type I error**: rejecting a true null. Probability α
- **Type II error**: failing to reject a false null. Probability β
- **Power**: the probability of rejecting a false null, $1 - \beta$

Errors and Power

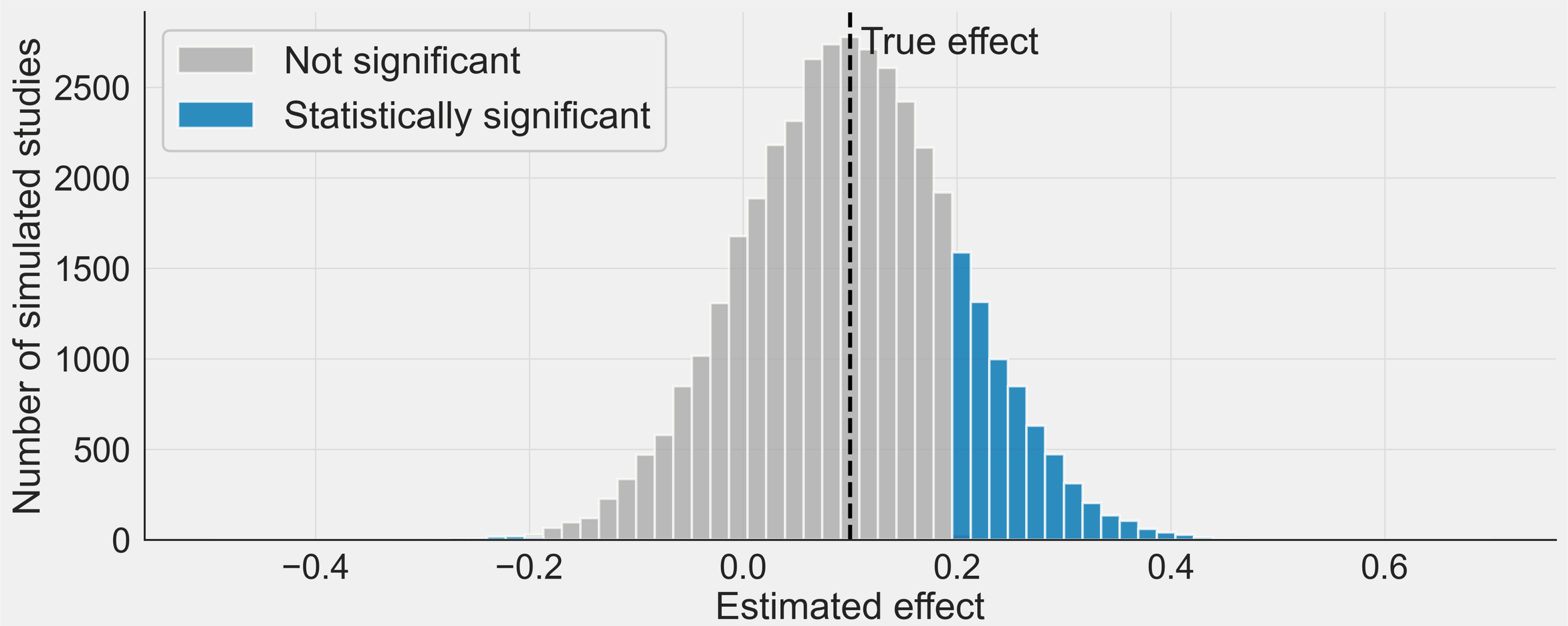


Sign and Magnitude Errors

Concepts introduced by Gelman and Carlin. - **Type S error**: the estimate has the wrong sign - **Type M error**: the estimate exaggerates the magnitude

Both are common when power is low.

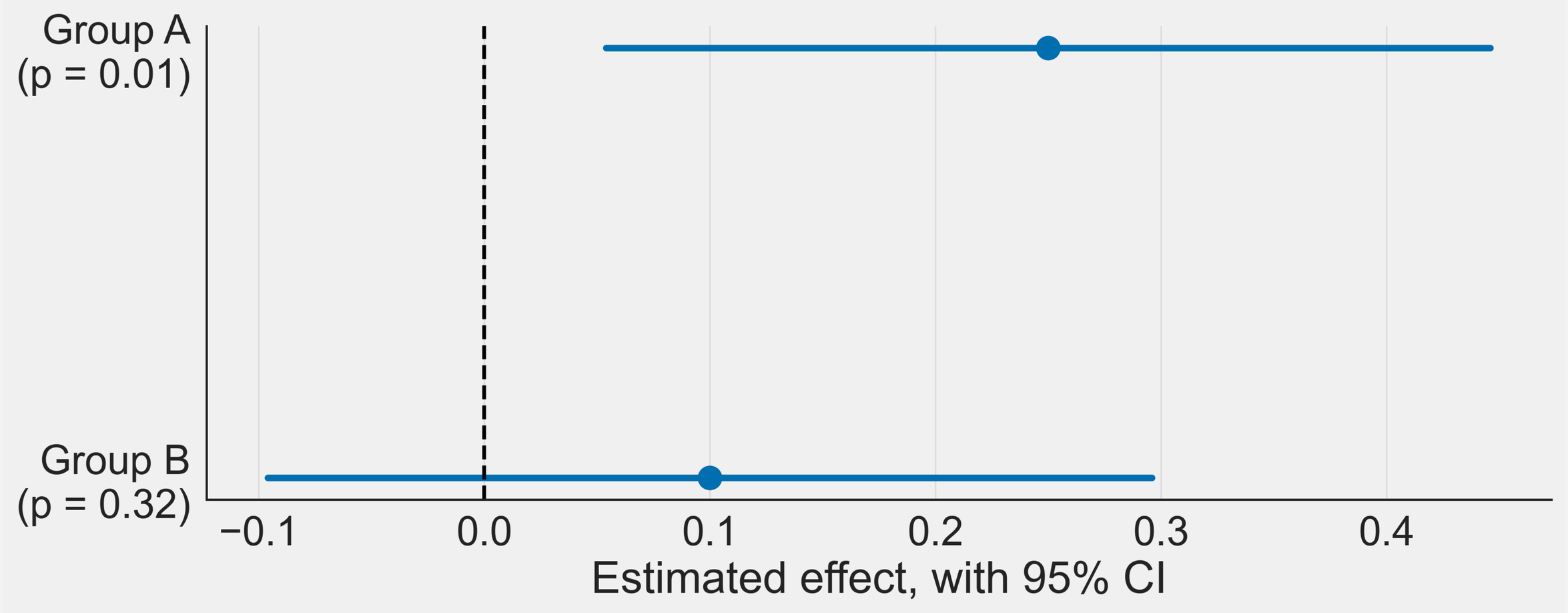
Significant Estimates Exaggerate



Non-Significance Is Not Zero

- Failing to reject is not evidence that the null is true
- A wide interval that includes zero is also compatible with large effects
- The difference between significant and not significant is not itself significant

Significant vs. Not Significant



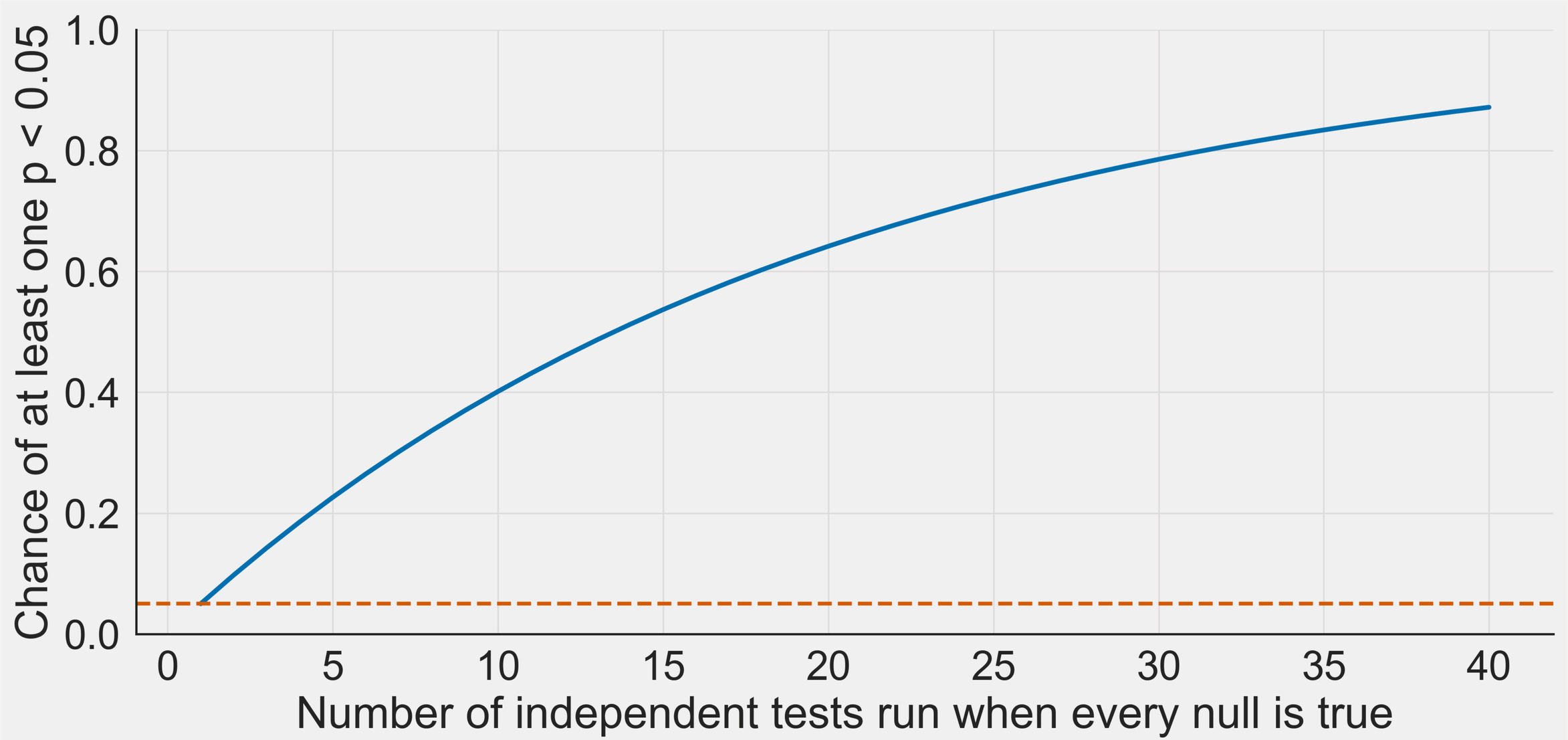
Researcher Degrees of Freedom

The Garden of Forking Paths

- Which observations to drop
- Which variables to control for
- How to code and transform variables
- Which outcome and which subgroup to report

Each choice is reasonable. Together they produce many possible analyses.

Many Tests, Many False Positives



Practicing Good Inference

- Define the estimand before looking at the data
- Preregister the analysis plan, or document deviations from it
- Run robustness checks and show them
- Report all of your outcomes, comparisons, and specifications
- Make your data and code public

Summary

- The estimand is the quantity of interest, stated before choosing a model, and the estimator is the rule used to estimate it
- A standard error measures sampling variability, and confidence intervals and test statistics are built from it
- Wald tests compare an estimate to a null value, and permutation tests build the null distribution from the data
- Chi-squared tests compare observed and expected counts, F-tests compare nested models, and Lagrange multiplier tests need only the restricted model
- A p-value does not tell you whether an effect is large, whether it is zero, or whether it is true, so report magnitudes and intervals and keep the analysis honest

Key Definitions

Key Definitions

- **Trial, outcome, event**: an action that produces a result, the result, and a set of results
- **Sample space**: the set of all possible outcomes of a trial
- **Random variable**: a number assigned to each outcome of a trial
- **PMF / PDF**: the probability a discrete variable takes a value / the height of the density curve for a continuous variable
- **CDF**: $P(X \leq x)$

Key Definitions

- **Law of large numbers**: the sample mean converges to the population mean as n grows
- **Central limit theorem**: the standardized sample mean approaches a standard normal as n grows
- **Parameter**: a fixed characteristic of a population
- **Estimand**: the precise quantity we want to estimate
- **Estimator / estimate**: the rule for computing a quantity from data / the number it produces

Key Definitions

- **Theoretical estimand**: the unit-specific quantity and target population, stated outside any model
- **Empirical estimand**: an observable quantity linked to the theoretical estimand by assumptions
- **Sampling distribution**: the distribution of an estimator across repeated samples
- **Standard error**: the standard deviation of the sampling distribution
- **Bootstrap**: estimating a standard error by resampling the data with replacement

Key Definitions

- **Z-score**: the distance from the mean in standard deviations
- **t-distribution**: a heavier-tailed distribution used when the standard error is estimated
- **Confidence interval**: a range built so that 95% of such intervals contain the true value across repeated samples
- **Test statistic**: a statistic computed from the data to test a null hypothesis
- **p-value**: the probability of a test statistic at least as extreme as the one observed, if the null is true

Key Definitions

- **NHST**: assume the null, then reject it if the data would be too surprising under it
- **Wald test**: compares an estimate to a null value in standard errors
- **Permutation test**: builds a null distribution by shuffling labels
- **Chi-squared distribution**: the distribution of a sum of d squared independent standard normal random variables, with d degrees of freedom
- **Contingency table**: counts for every combination of two categorical variables

Key Definitions

- **Expected count**: the count in a cell if the variables are independent, (row total × column total) / n
- **Pearson chi-squared test**: tests independence using $\sum(O-E)^2/E$, with $(r-1)(c-1)$ degrees of freedom
- **Goodness-of-fit test**: compares the counts of one variable to hypothesized proportions
- **R-squared**: the share of the variance in the outcome explained by the model, $1 - \text{SSE}/\text{SST}$
- **SST**: the total sum of squares, the SSE of the naive model that predicts $\bar{Y}$

Key Definitions

- **F-test**: compares the fit of a restricted and an unrestricted model
- **Restricted / unrestricted model**: the model with q restrictions imposed / the model without them
- **F-distribution**: the distribution of the F statistic under the null, with q and $n - k$ degrees of freedom
- **Nested / non-nested models**: one model is a special case of the other / neither is
- **AIC / BIC**: fit plus a penalty for the number of coefficients; lower is better

Key Definitions

- **Lagrange multiplier (score) test**: tests a restriction using the slope of the log-likelihood at the restricted estimate
- **Type I / Type II error**: rejecting a true null / failing to reject a false null
- **Power**: the probability of rejecting a false null
- **Type S / Type M error**: a wrong-signed estimate / an exaggerated estimate
- **Garden of forking paths**: many defensible analysis choices that depend on the data